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GREGORY J. SCHWING

EDUCATION

2018-present **Wayne State University School of Medicine**
Detroit, Michigan
MD anticipated June 2026

2018-present **Wayne State University School of Engineering**
Detroit, Michigan
PhD (Computer Science) anticipated June 2026

2016-2018 **University of New Orleans**
New Orleans, Louisiana
BS Computer Science

2011-2015 **University of New Orleans**
New Orleans, Louisiana
BS Biological Science

HONORS/AWARDS

Medical School:

2024 Clerkship Honors – Obstetrics/Gynecology, Family Medicine
2022-present NIH Ruth L. Kirschstein Individual Predoctoral Fellowship (F30)
2022 Wayne State University Extramural Funding Incentive Award
2022-2023 Rumble Fellowship (Declined)
2018-2020 Board of Governors Full Tuition Scholarship, Wayne State University School of Medicine

Undergraduate:

2018 ACERS Scholarship (Achievement in ComputER Science)
2018 Departmental Honors in Computer Science (Thesis-option)
2018 Magna Cum Laude Graduate
2015 Academic Excellence Award
2015 Biological Sciences Achievement Award
2015 Magna Cum Laude Graduate

LICENSURE/CERTIFICATION

2020 **USMLE** (identification number (5-436-197-7)
Step 1 – Pass
Step 2 – 259
2024 ACLS certified through September 2026

PROFESSIONAL MEMBERSHIPS/SOCIETIES

2024-present American Medical Association, Member
2024-present Association for Computing Machinery, Member
2019-present American Association of Neurological Surgeons, Member

WORK/VOLUNTEER EXPERIENCE

2024-present Catholic Medical Student Association, President

2024 Robert R. Frank Student Run Free Clinic, Detroit, MI

2018-2019 Auntie Na's Clinic, Detroit, MI

2019 Child Life, Children's Hospital of Michigan, Detroit, MI

2018 Gleaner's Food Bank, Detroit, MI

2018 Robert R. Frank Student Run Free Clinic

2018 Detroit Community Clean Up

2016-2017 Child Life, Children's Hospital, New Orleans, LA

2015 Happy Cart Volunteer, St Jude Children's Research Hospital. Memphis, TN

2015 Emergency Room Scribe, Ochsner Hospital, Metairie, LA

2013-2015 Caregiver / Nurse Assistant, Metairie, LA

2013-2014 Patient Transporter / Lift Team, East Jefferson General Hospital, Metairie, LA

2013-2014 Warehouse Employee, Jazzmen Rice, Metairie, LA

2012 Habitat for Humanity, Mobile, AL

2006-2015 Cashier / Cook / Assistant Manager, Lee's Hamburger Restaurant, Metairie, LA

TEACHING EXPERIENCE

2019-2020 ALS Case Based Learning Facilitator—First Year Wayne State SOM Medical Students

2019-2020 Case Based Learning Facilitator—First Year Wayne State SOM Medical Students

2018-2019 Auntie Na's Village Literacy teacher—early school-age children

2018 In Silico Computation Biology Seminar Facilitator—Graduate students

2011-2012 Mobile Alabama Public School Tutor—middle school age children

RESEARCH EXPERIENCE

2018-2024 Schwiebert-Potoff Lab, https://gomc-wsu.org/
Wayne State University, Detroit MI
MD-PhD, Computer Science/Chemical Engineering
- C++ development
- Hybrid Monte-Carlo/Molecular Dynamics Methods development
- Python scripting

2017-2018 Summa Lab, https://www.cs.uno.edu/~csumma
Louisiana State University Health Sciences Center-New Orleans, New Orleans LA
Summer Research Fellow, Computer Science/Computational Structural Biochemistry
- C++ development
- Molecular Dynamics Methods development
- Python scripting

2017 Hicks Lab, https://www.medschool.lsuhsc.edu/bioinformatics/
Louisiana State University Health Sciences Center-New Orleans, New Orleans LA
Summer Research Fellow, Bioinformatics
- Data mining
- Differential Gene Expression analysis
- INGENUITY Gene Pathway analysis

2016 Basu Ray Lab, https://ibasuray.com/
Tulane Hospital, New Orleans, LA
Clinical Research Associate, Cardiology
- Chart Review
- SAS Statistical Analysis
- Case-control studies

2015 Gawad Lab, https://gawadlab.org/
St. Jude Children's Research Hospital, Memphis TN
Pediatric Oncology Education Fellow, Oncology
- Plasmid vector constructs planning and creation
- Single-cell Sequencing
- Cellular barcoding
- Cell culture

2014 Liu Lab, https://liulabatuno.github.io/
University of New Orleans, New Orleans LA
Undergraduate Research Assistant, Biochemistry
- Plasmid vector constructs planning and creation
- Protein concentration spectrometry
- Gel electrophoresis

GRANTS AND CONTRACT AWARDS

2022-present Myc Transcription Factor Inhibitor Design: Integrating Atomic and Mesoscale with Semi-Supervised Generative Deep Learning Models

Sponsor: National Institute of General Medical Sciences
Sponsor Award Number: 1F30GM147909-01
Role: Principal Investigator, 100% Effort
Total Direct Budget: $133,350.27

THESES/DISSERTATIONS

Schwing, Gregory John, “Matchings and Molecules: Ball and Stick Models for Cheminformatics,” February 2024, Ph.D. Thesis

Schwing, Gregory John, "Implementation of Replica Exchange with Dynamic Scaling in GROMACS 2018," May 2018, Senior Honors Thesis.

ORAL PRESENTATIONS

Schwing, Gregory, Grosu Daniel, Schwiebert, Loren, Shared-Memory Parallel Edmonds Blossom Algorithm for Maximum Cardinality Matching in General Graphs. 2024 International Parallel and Distributed Processing Symposium-Workshops, May 27, 2024 – May 31, 2024, San Francisco, CA

Schwing, Gregory, Grosu Daniel, Schwiebert, Loren, Parallel Maximum Cardinality Matching for General Graphs on GPUs. 2024 International Parallel and Distributed Processing Symposium-Workshops, May 27, 2024 – May 31, 2024, San Francisco, CA

Schwing, Gregory, Barhaghi, Mohammad Soroush, Crawford, Brad, Schwiebert, Loren, Potoff, Jeffrey, Leveraging the Wolf Method for Electrostatics to Extend Time and Length Scales Accessible By Monte Carlo Simulations, 2022 AIChE Annual Meeting, 13-18 November 2022, Phoenix, Arizona

Crawford, Brad, Barhaghi, Mohammad Soroush, **Schwing, Gregory**, Hardy, David, Stone, John, Schwiebert, Loren, Potoff, Jeffrey, Tajkhorshid, Emad, py-MCMD: Hybrid Monte Carlo–Molecular Dynamics Simulations, 2022 AIChE Annual Meeting, 13-18 November 2022, Phoenix, Arizona

Crawford, Brad, Barhaghi, Mohammad Soroush, **Schwing, Gregory**, Hardy, David , Stone, John, Schwiebert, Loren, Potoff, Jeffrey, Tajkhorshid, Emad, Hybrid Monte Carlo–Molecular Dynamics Simulations, 2020 Virtual AIChE Annual Meeting, 15-19 November 2021, Virtual

Nejahi, Younes, **Schwing, Gregory**, Schwiebert, Loren, Potoff, Jeffrey, Unsupervised Optimization of Non-Bonded Parameters in Molecular Mechanics Force Fields Using the Particle Swarm Method, 2020 Virtual AIChE Annual Meeting, 16-20 November 2020, Virtual

Schwing, G., Moisi, M. and Hicks, C., 2019, July. A Novel Bioinformatics Approach to Estimate Hazard in Pediatric High Grade Glioma. 2019 American Association of Neurological Surgeons Scientific Sessions, April 13-17, 2019, San Diego

POSTER PRESENTATIONS

Nejahi, Younes, Barhaghi, Mohammad Soroush, **Schwing, Gregory**, Schwiebert, Loren, Potoff, Jeffrey, GPU Optimized Monte Carlo Version 2.50, 2020 Virtual AIChE Annual Meeting, 16-20 November 2020, Virtual

Ray, I.B., **Schwing, G.**, Novak, J., Monlezun, D.J., Boja, H., Deere, B., Duque, J., Park, M., Lawrence, J., Middour, T., Morris, S., Boulad, A., Carry, B., Abi Rafey, N., Ali, G., Anwar, A., LeJemtel, T., Martin-Schild, S., & Cassidy, M. (Nov, 2016). Complex left atrial appendage morphology increases risk of cryptogenic stroke Poster presented at: American Heart Association Scientific Sessions and Resuscitation Science Symposium; New Orleans, LA, USA.

Ray, I.B., **Schwing, G.**, Novak, J., Monlezun, D.J., Boja, H., Deere, B., Duque, J., Park, M., Lawrence, J., Middour, T., Morris, S., Boulad, A., Carry, B., Abi Rafey, N., Ali, G., Anwar, A., LeJemtel, T., Martin-Schild, S., & Cassidy, M. (Jan, 2017). Complex left atrial appendage morphology is an independent risk factor for cryptogenic stroke Poster presented at: Atrial Fibrillation 22nd Annual Symposium; Orlando, FL, USA.

Ray, I.B., **Schwing, G.**, Middour, T., Monlezun, D.J., Katikitala, K., Koranne, K., Cassidy, M., Martin-Schild, S., Lam, W., Afsar, H., Giorgberidze, I., Cheng, J., Saeed, M. (Mar, 2017). Significant diastolic dysfunction is associated with cardioembolic stroke independent of history of atrial fibrillation Poster presented at: American College of Cardiology Annual Scientific Session & Expo; Washington, DC, USA.

Ray, I.B., **Schwing, G.**, Middour, T., Monlezun, D.J., Katikitala, K., Koranne, K., Martin-Schild, S., Lam, W., Afsar, H., Giorgberidze, I., Cheng, J., Saeed, M. (Apr, 2017). Significant diastolic dysfunction is associated with cardioembolic stroke independent of history of atrial fibrillation Poster presented at: 74th World Congress on Acute Heart Failure; Paris, France.

Ray, I.B., **Schwing, G.**, Middour, T., Monlezun, D.J., Katikitala, K., Koranne, K., Martin-Schild, S., Lam, W., Afsar, H., Giorgberidze, I., Cheng, J., Saeed, M. (Aug, 2017). Heart failure with preserved ejection fraction is associated with cardioembolic stroke independent of history of atrial fibrillation Poster presented at: European Society of Cardiology World Congress; Barcelona, Spain.

RESEARCH PUBLICATIONS

Schwing, G., Grosu, D. and Schwiebert, L., Shared-Memory Parallel Edmonds Blossom Algorithm for Maximum Cardinality Matching in General Graphs, In *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. 2024

Schwing, G., Grosu, D. and Schwiebert, L., Parallel Maximum Cardinality Matching for General Graphs on GPUs. In *2024 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*. 2024

Barhaghi, M.S., Crawford, B**., Schwing, G.**, Hardy, D.J., Stone, J.E., Schwiebert, L., Potoff, J. and Tajkhorshid, E., Py-MCMD: python software for performing hybrid Monte Carlo/molecular dynamics simulations with GOMC and NAMD, Journal of chemical theory and computation, 2022

Schwing, G., Palese, L.L., Fernández, A., Schwiebert, L. and Gatti, D.L., 2022. Molecular dynamics without molecules: searching the conformational space of proteins with generative neural networks. arXiv preprint arXiv:2206.04683.

Nejahi, Y., Barhaghi, M.S., **Schwing, G.**, Schwiebert, L. and Potoff, J., Update 2.70 to "GOMC: GPU optimized Monte Carlo for the simulation of phase equilibria and physical properties of complex fluid, SoftwareX, 2021

Rick, S.W., **Schwing, G.J.** and Summa, C.M.,, An implementation of replica exchange with dynamical scaling for efficient large-scale simulations. Journal of Chemical Information and Modeling, 2021

Hoerger, M., Wayser, G.R., **Schwing, G.**, Suzuki, A. and Perry, L.M.,, Impact of interdisciplinary outpatient specialty palliative care on survival and quality of life in adults with advanced cancer: a meta-analysis of randomized controlled trials, Annals of Behavioral Medicine, 2019

Basu-Ray, I., Sudhakar, D., **Schwing, G.,** Monlezun, D., Zhang, L., Shah, S.K., Pujara, D., Ting, K., Rafeh, N.A., Ali, G. and Cassidy, M., Complex left atrial appendage morphology is an independent risk factor for cryptogenic ischemic stroke, Frontiers in cardiovascular medicine, 2018

INTERESTS/EXTRACURRICULAR ACTIVITIES

Classical Pianist
Marathon Hiker (Rim2Rim – Grand Canyon)
Lead Altar Server (Catholic)
High School Wrestler, 4-year letterman
Competitive chess player
Weightlifting
Literature (Dostoyevsky, Goethe)
Psychology (Jung, Peterson)